

NAZIM HUSSAIN

AI/ML Engineer | LLM & RAG Specialist | Full-Stack AI Developer

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SUMMARY

Final-year BS Artificial Intelligence student (2022–2026, Quaid-e-Awam University) and AI/ML Engineer specializing in Large Language Models, Retrieval-Augmented Generation (RAG), and full-stack AI deployment. Built and publicly deployed a bilingual English/Urdu legal RAG pipeline indexing 500+ documents using hybrid FAISS + BM25 retrieval, cross-encoder re-ranking, and LangChain, achieving sub-2-second response latency on Hugging Face Spaces. Additional projects include a CNN-based digit recognition system (TensorFlow), a computer-vision trading bot built with OpenCV, and a FaceNet face-recognition system running at 30 FPS. Proficient in Python, PyTorch, TensorFlow, LangChain, LangGraph, Flask, FastAPI, and Docker, with hands-on MLOps experience. Seeking AI/ML Engineer, LLM Engineer, or Associate Software Engineer roles.

SKILLS

AI / ML: Python, PyTorch, TensorFlow, Deep Learning, NLP

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), SQL

LLM & RAG: LangChain, LangGraph, Hugging Face, FAISS, Prompt Engineering

Backend & Deployment: Flask, FastAPI, Docker, REST APIs

Computer Vision: OpenCV, YOLOv8

Tools: Git, GitHub, VS Code

Soft Skills: Quick Learner, Strong Communication, Attention to Detail, Deadline-Driven, Team Collaboration

EXPERIENCE

Computer Operator

Jan 2022 – Jan 2023

Citizen Police Liaison Committee (CPLC), Shaheed Benazirabad

- Managed and maintained computer systems and organizational databases, ensuring data integrity and system reliability
- Supported data entry, verification, and reporting workflows across departments
- Troubleshoot software and hardware issues, minimizing downtime and improving team efficiency

PROJECTS

Pakistan Legal Advisor | *LangChain, FAISS, BM25, Cross-Encoder Re-ranking, LLM Integration, Hugging Face*

- Built a multilingual (Urdu & English) legal assistant on a production RAG pipeline indexing 500+ Pakistani legal documents
- Architecture: PDF ingestion → recursive text chunking → dual retrieval (FAISS dense vectors + BM25 sparse) → cross-encoder re-ranking → LLM answer generation with source citation
- Achieved sub-2-second end-to-end response latency; deployed publicly on Hugging Face Spaces

Digit Recognition System | *CNN, TensorFlow, MNIST, Flask, Data Augmentation*

- Trained a CNN on 60,000 MNIST samples achieving 92% test accuracy
- Applied dropout regularization (rate 0.5) and real-time data augmentation (rotation, zoom, shift), reducing overfitting by ~4% vs. baseline
- Deployed via Flask with a canvas-based web UI for live digit prediction, averaging <100ms inference per request

AI-Powered Trading Bot | *OpenCV, Python Automation, Algorithmic Trading, Backtesting*

- Built an automated trading system that processes real-time OHLCV candlestick charts using OpenCV (Canny edge detection, contour analysis) to identify market structure breaks, volume spikes, and retest zones
- Generated timestamped buy/sell signals with configurable risk parameters; backtested across 6 months of historical data with 62% signal accuracy on trending markets

Face Recognition System | *OpenCV, FaceNet, Edge Deployment, Cosine Similarity*

- Built a real-time face detection and recognition pipeline using OpenCV (Haar cascades + DNN detector) and FaceNet 512-d embeddings with cosine similarity matching
- Achieved 30 FPS on standard hardware and ~95% recognition accuracy across a 50-person database, with a sub-200ms identification pipeline optimized for edge devices

EDUCATION

BS Artificial Intelligence, GPA 3.4/4.0

2022 – 2026 (*Expected*)

Quaid-e-Awam University of Engineering, Sciences & Technology

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Neural Networks, Transformers, NLP, Computer Vision, MLOps, Optimization Techniques
- Built end-to-end AI projects: multilingual RAG Legal Advisor, real-time Face Recognition System, CNN Digit Recognizer, AI Trading Bot
- Active on Kaggle with 9 badges, applying classroom concepts to real-world datasets

CERTIFICATIONS

- Generative AI with Capstone Project, Coursera (2024)
- AI Essentials, Coursera (2024)